

1 Solution — GIAM 8.1

Problem 2

1.1 Problem

Prove that set equivalence is an equivalence relation.

1.2 Setup — Definitions

Section 8.1 defines, for any two sets A and B ,

$$A \equiv B \iff \text{there exists a bijection } f : A \rightarrow B.$$

Recall that $f : A \rightarrow B$ is a *bijection* iff it is both

- *injective* (one-to-one): for all $x_1, x_2 \in A$, $f(x_1) = f(x_2) \implies x_1 = x_2$; and
- *surjective* (onto): for every $b \in B$, there exists $a \in A$ with $f(a) = b$.

To show $\equiv$ is an equivalence relation on (the class of) sets, we must verify three properties:

- **Reflexive:** $A \equiv A$ for every set A .
- **Symmetric:** $A \equiv B \implies B \equiv A$.
- **Transitive:** $A \equiv B$ and $B \equiv C \implies A \equiv C$.

We prove each in turn.

1.3 Reflexivity

Claim. For every set A , $A \equiv A$.

Proof. Consider the **identity function** $\text{id}_A : A \rightarrow A$ defined by $\text{id}_A(x) = x$ for all $x \in A$.

- *Injective:* Suppose $\text{id}_A(x_1) = \text{id}_A(x_2)$. Then $x_1 = x_2$ directly from the definition. So id_A is injective.

- *Surjective*: For any $a \in A$ (in the codomain), take $a \in A$ (in the domain). Then $\text{id}_A(a) = a$. So id_A is surjective.

Hence id_A is a bijection from A to A , witnessing $A \equiv A$. ■

1.4 Symmetry

Claim. For all sets A, B , if $A \equiv B$ then $B \equiv A$.

Proof. Assume $A \equiv B$. By definition, there is a bijection $f : A \rightarrow B$. We must produce a bijection $g : B \rightarrow A$.

Consider the **inverse relation** $f^{-1} \subseteq B \times A$ defined by

$$(b, a) \in f^{-1} \iff (a, b) \in f \iff f(a) = b.$$

We show f^{-1} is a function from B to A , and that it is bijective.

- f^{-1} is *total* (every $b \in B$ has at least one image): since f is *surjective*, for each $b \in B$ there is some $a \in A$ with $f(a) = b$, hence $(b, a) \in f^{-1}$.
- f^{-1} is *well-defined* (each b has at most one image): suppose $(b, a_1) \in f^{-1}$ and $(b, a_2) \in f^{-1}$. Then $f(a_1) = b = f(a_2)$, and since f is *injective*, $a_1 = a_2$.

So f^{-1} is a function from B to A . Write $g = f^{-1}$.

Now verify g is itself a bijection.

- g is *injective*: suppose $g(b_1) = g(b_2) = a$. By definition of g , this means $f(a) = b_1$ and $f(a) = b_2$. Since f is a function, $b_1 = b_2$.
- g is *surjective*: given $a \in A$, let $b = f(a) \in B$. Then $g(b) = a$ by definition of g .

Hence $g : B \rightarrow A$ is a bijection, witnessing $B \equiv A$. ■

Remark. The textbook's outline observed that " f^{-1} is a function" follows from f being injective, and " f^{-1} is one-to-one" follows from f being a function. We split this a bit differently for clarity, but the content is the same: each property of f gives one of the two halves needed for the other property of f^{-1} .

1.5 Transitivity

Claim. For all sets A, B, C , if $A \equiv B$ and $B \equiv C$, then $A \equiv C$.

Proof. Assume $A \equiv B$ and $B \equiv C$. So there exist bijections $f : A \rightarrow B$ and $g : B \rightarrow C$.

Consider the **composition** $h = g \circ f : A \rightarrow C$ defined by

$$h(a) = (g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A.$$

We verify h is a bijection.

- *h is injective:* suppose $h(a_1) = h(a_2)$, i.e., $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$.
- *h is surjective:* let $c \in C$. Since g is surjective, there is $b \in B$ with $g(b) = c$. Since f is surjective, there is $a \in A$ with $f(a) = b$. Then $h(a) = g(f(a)) = g(b) = c$.

Hence $h : A \rightarrow C$ is a bijection, witnessing $A \equiv C$. ■

1.6 Conclusion

The relation $\equiv$ is reflexive, symmetric, and transitive, so $\equiv$ is an equivalence relation on the class of sets. ■

1.7 Summary Table — Which Property of f Does What

It is illuminating to track *which* property of f (or g) is used at each step:

Property of $\equiv$	Witness	Property of f used
reflexive	id_A	(none — identity is automatically bijective)
symmetric	f^{-1}	f surjective $\Rightarrow f^{-1}$ total
		f injective $\Rightarrow f^{-1}$ well-defined
transitive	$g \circ f$	f, g injective $\Rightarrow g \circ f$ injective
		f, g surjective $\Rightarrow g \circ f$ surjective

Table 1.1

In short:

- **Reflexivity** is free, because every set comes with the identity map.
- **Symmetry** is the statement “*bijections can be inverted.*”
- **Transitivity** is the statement “*bijections compose.*”

These three facts about bijections are precisely what make “there is a bijection” a sensible notion of *sameness* between sets.

1.8 Remark — Why Not Just Define $A \equiv B$ by $|A| = |B|$?

The textbook flags the circularity of the would-be definition $A \equiv B \Leftrightarrow |A| = |B|$: we are trying to *define* the cardinalities $|A|$ as the equivalence classes of $\equiv$, so we cannot use $|A|$ to set up $\equiv$ in the first place.

The bijection-based definition sidesteps this entirely. A bijection $f : A \rightarrow B$ is a well-defined object — a set of ordered pairs $(a, b) \in A \times B$ satisfying functional and bijective conditions — that we can talk about *without ever mentioning cardinality*. Once $\equiv$ is an equivalence relation (which is what this problem establishes), we are *then* entitled to *define* “ $|A|$ ” as the equivalence class of A under $\equiv$. This is what Section 8.1 quietly does when it identifies the finite cardinalities with the classes of $\emptyset, \mathbb{N}_1, \mathbb{N}_2, \dots$.

1.9 Remark — A Subtle Foundational Point

Strictly speaking, $\equiv$ is not a relation on a *set* (there is no “set of all sets” in standard ZFC set theory — that would lead to Russell’s paradox). Rather, $\equiv$ is a relation on the *proper class* of all sets. The three properties we verified (reflexive, symmetric, transitive) all make sense at the class level — the definitions involve only quantification over the sets being compared — so the proof above goes through unchanged. The textbook elides this distinction, which is fine for the purposes of Chapter 8; the formal treatment lives in axiomatic set theory and is not needed here.